Practice Problems
Calculus-Based Physics II

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Chapter 36 Part B Electric Fields

1) A uniform conducting wire has a linear charge density of $\lambda = 1 \times 10^{-5}$ C/m, determine the total charge on a 0.25 m length of the wire.

$\lambda = \frac{Q}{L}$ | $Q = \lambda L = 2.5 \times 10^{-6}$ C

2) A uniform conduction wire has a total charge of 0.02 C and a linear charge density of $\lambda = 1 \times 10^{-5}$ C/m, determine the length of the wire.

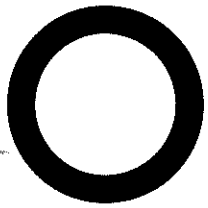
$\lambda = \frac{Q}{L}$ | $L = \frac{Q}{\lambda} = 0.2$m

3) A circular conducting disk has a total charge of 2×10^{-6} C and a surface charge density of $\sigma = 5 \times 10^{-6}$ C/m^2, determine the radius of the disk.

$\sigma = \frac{Q}{A} = \frac{Q}{\pi r^2}$ | $r = \sqrt{\frac{Q}{\pi \sigma}} = 0.36$m

4) A conducting sphere with an outside radius of $r_{out} = 0.5$ m and an inside radius of $r_{in} = 0.4$ m has a net negative charge of positive $q = 8$ μC on the outside. Determine the surface charge density on the outside of the sphere?

$\sigma = \frac{Q}{A} = \frac{Q}{4\pi r^2} = 2.5 \times 10^{-6}$ C/m^2

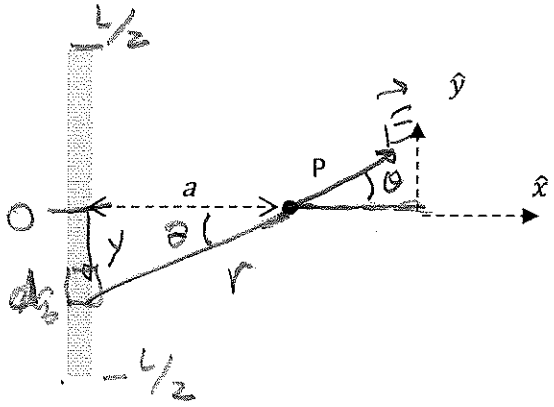


5) The volume charge density is given as rho $\rho = Q/V$, where Q is the total charge and V is the volume the charge is in. Determine the total charge of a sphere of radius 5 cm if it has a total uniform charge throughout the volume with a volume charge density of 7×10^{-5} C/m^3.

$\rho = \frac{Q}{V}$ | $Q = \rho V = \rho\left(\frac{4}{3}\pi r^3\right)$

$Q = 3.66 \times 10^{-8}$ C

6) A positively charged straight wire with a uniform linear charge density of λ, determine the electric field strength at point P, "a" meters above the wire and situated at the above the center of the wire of length L. Assume that a and L are constant and do not change in this problem.



A) Draw and label on the diagram the incremental charge, radius, angle, electric Field and any other variables needed to solve the problem.

B) Determine the incremental charge on the wire.

$$dq = \lambda dl = \lambda dy$$

C) Determine the x component of the electric field strength

$$E_x = \vec{E}\cdot\hat{x} = |\vec{E}|\cos(\alpha) = |\vec{E}|\cos(\theta) = K\int \frac{\cos\theta\, dq}{r^2}$$

$$r^2 = y^2 + a^2 \qquad \cos\theta = \frac{a}{r} = \frac{a}{(y^2+a^2)}$$

$$E_x = K\int \frac{\cos\theta\,\lambda\, dy}{r^2} = K\lambda\int \frac{a}{(y^2+a^2)}\cdot\frac{1}{(y^2+a^2)^{3/2}}\,dy$$

$$E_x = K\lambda a\int_{-L/2}^{L/2}\frac{1}{(y^2+a^2)^{3/2}}\,dy = K\lambda a\left[\frac{y}{a^2(y^2+a^2)^{1/2}}\right]_{-L/2}^{L/2}$$

$$E_x = \frac{a K\lambda}{a^2}\left[\frac{(L/2)}{((L/2)^2+a^2)^{1/2}} - \frac{-(L/2)}{((L/2)^2+a^2)^{1/2}}\right]$$

D) Determine the y component of the electric field strength

$$E_x = \frac{K\lambda}{a}\left[\frac{L}{((L/2)^2+a^2)^{1/2}}\right]$$

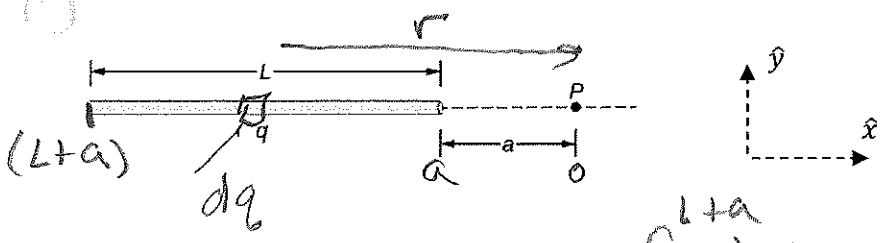
$$E_y = \vec{E}\cdot\hat{y} = |\vec{E}|\cos(\beta)$$

$$E_y = |\vec{E}|\cos(90^\circ-\theta) = |\vec{E}|\sin\theta \quad \Big| \quad \sin\theta = \frac{y}{r}$$

$$E_y = K\int \frac{\sin\theta\,\lambda\, dy}{r^2} = K\lambda\int_{-L/2}^{L/2}\frac{y}{(y^2+a^2)^{1/2}}\frac{1}{(y^2+a^2)}\,dy$$

$$E_y = K\lambda\int \frac{y}{(y^2+a^2)^{3/2}}\,dy = K\lambda\left[\frac{1}{(y^2+a^2)^{1/2}}\right]_{-L/2}^{L/2} = 0$$

7) A Thin wire of length L with a uniform linear charge density of λ is shown below, determine the electric field strength along the x-axis at point P due to the charge on the wire. Draw and label on the diagram the incremental charge, radius, angle, electric Field and any other variables needed to solve the problem.



$r = x$

$dq = \lambda dx$

$$E_x = k\int \frac{dq}{r^2} = k\int_a^{L+a} \frac{\lambda dx}{x^2} = -k\lambda\left[\frac{1}{x}\right]_a^{L+a}$$

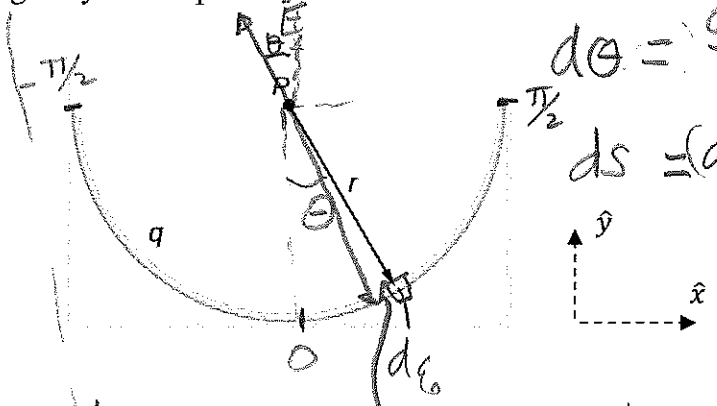
$$E_x = -k\lambda\left[\frac{1}{L+a} - \frac{1}{a}\right]$$

8) A semicircular thin wire with a uniform ~~linear~~ charge density of λ,

A) Draw and label on the diagram the incremental charge, radius, angle, electric Field and any other variables needed to solve the problem.

B) Determine the electric field strength along the y axis at point P, which is at the radius of the semicircle.

Changing θ
r is Constant



$d\theta = \frac{ds}{r}$

$ds = (d\theta)r$

$E_y = \vec{E}\cdot\hat{y} = |\vec{E}|\cos(\beta)$

$E_y = |\vec{E}|\cos(\theta)$

$dq = r\,d\theta$

$$E_y = k\int \frac{\cos(\theta)\,dq}{r^2} = \frac{k\lambda}{r^2}\int_{-\pi/2}^{\pi/2} \cos\theta\, r\, d\theta$$

$$E_y = \frac{k\lambda}{r}\int_{-\pi/2}^{\pi/2} \cos\theta\, d\theta = \frac{k\lambda}{r}\left[\sin(\pi/2) - \sin(-\pi/2)\right] = \frac{2k\lambda}{r}$$

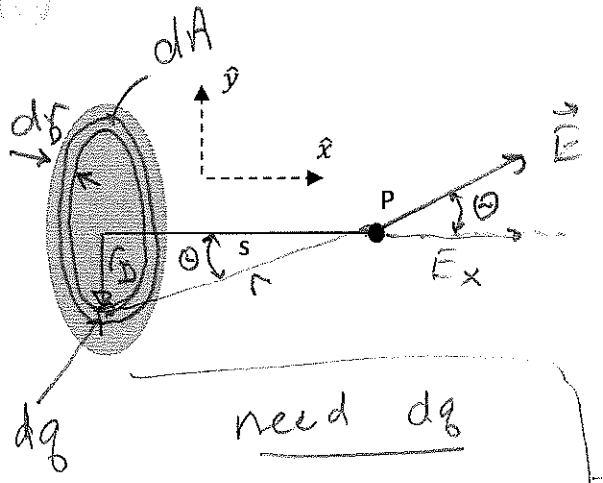
C) Show that the electric field strength along the x axis at point P is equal to zero.

$E_x = \vec{E}\cdot\hat{x} = |\vec{E}|\cos(90°+\theta) = |\vec{E}|\sin(\theta)$

$$E_x = k\int \frac{\sin\theta}{r^2}\,dq = \frac{k\lambda}{r}\int_{-\pi/2}^{\pi/2} \sin(\theta)\,d\theta$$

$$E_x = \frac{k\lambda}{r}\left[\cos(\pi/2) - \cos(-\pi/2)\right] = 0$$

9) A circular disk with a radius R and a uniform surface charge density of σ is shown below, determine the electric field at point P along the x-axis which is at a distance of s from the center of the disk. Draw and label on the diagram the incremental charge, radius, angle, electric Field and any other variables needed to solve the problem.



$\hat{x}$

$\vec{E} \cdot \hat{x} = |\vec{E}| \cos(\theta) = E_x$

$E_x = K \int \frac{dq}{r^2}$

need dq

r_D → radius of incremental disk

r → distance from dq to point P

R → radius of disk

$dq = \sigma\, dA$

dA - area of incremental disk

dA = area of ring

$dA = (C)\, dr$ (C: circumference)

$dA = (2\pi r_D)\, dr_D$

$dq = \sigma\, dA = \sigma (2\pi r_D)\, dr_D$

need variables in terms of r

$r^2 = s^2 + r_D^2$

$\cos\theta = \frac{s}{r} = \frac{s}{(s^2 + r_D^2)^{1/2}}$

$E_x = K \int \frac{dq}{r^2} = K \int \frac{\cos\theta (\sigma 2\pi r_D)\, dr_D}{r^2}$

$E_x = 2\pi K \sigma \int \left(\frac{s}{(s^2 + r_D^2)^{1/2}}\right)\left(\frac{1}{(s^2 + r_D^2)}\right) r_D\, dr_D$

$E_x = s\, 2\pi \sigma K \int_0^R \frac{r_D}{(s^2 + r_D^2)^{3/2}}\, dr_D$

look up integral

$E_x = s K \sigma 2\pi \left[\frac{1}{(r_D^2 + s^2)^{1/2}}\right]_0^R$

$E_x = s K \sigma 2\pi \left[\frac{1}{(R^2 + s^2)^{1/2}} - \frac{1}{s}\right]$

Answers

1) 2.5×10^{-6} C
2) 0.2 m
3) 0.36 m
4) 2.5×10^{-6} C
5) 3.66×10^{-8} C
6) See solutions
7) See solutions
8) See solutions
9) See solutions